PROJECT REPORT

TO-DO LIST APPLICATION USING SCRATCH

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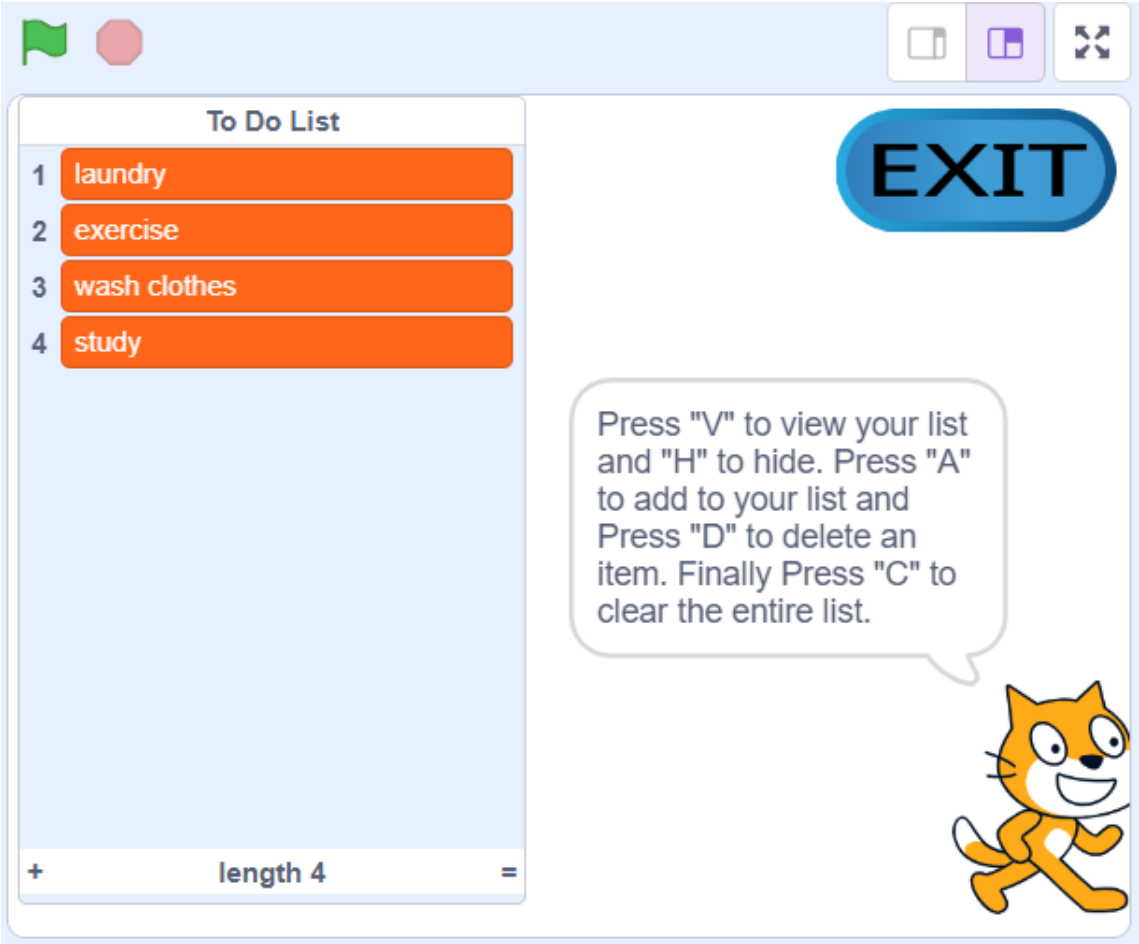
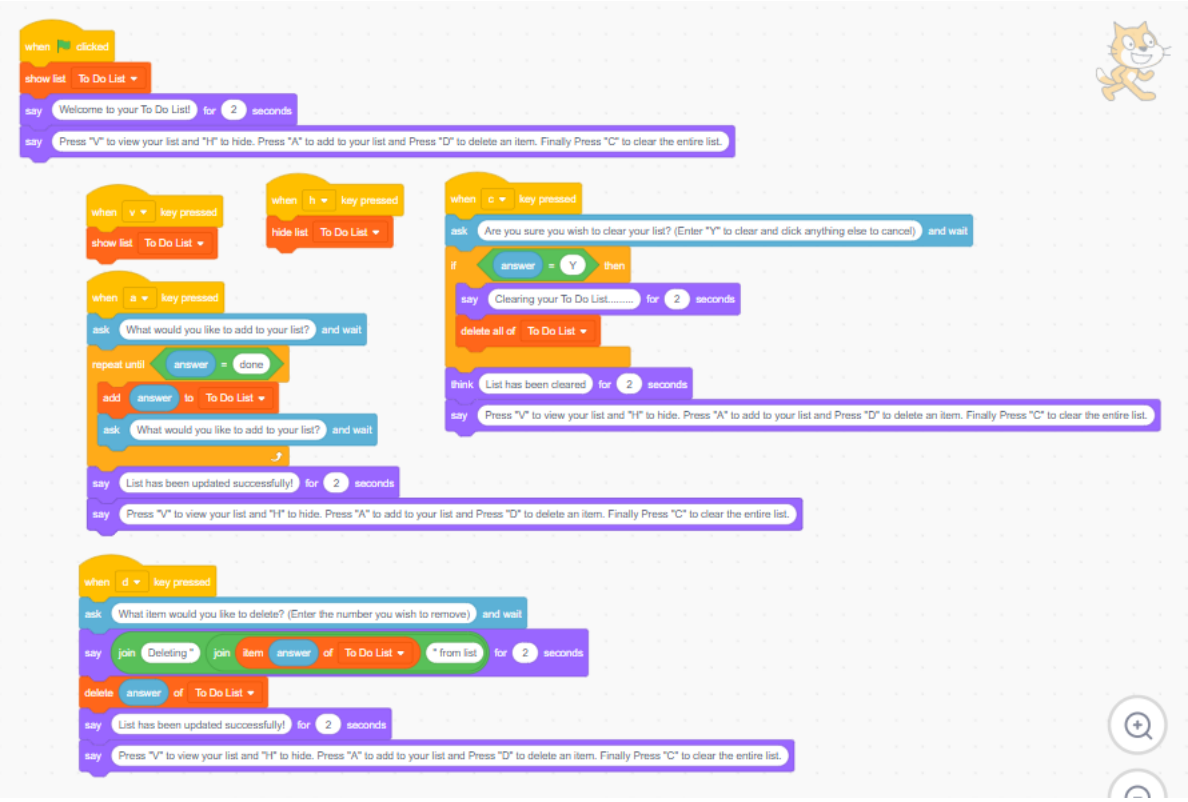
1. INTRODUCTION

This project is a simple and interactive **To-Do List Application** developed using Scratch. The purpose of this project is to help users manage their daily tasks efficiently by adding, viewing, deleting, hiding, and clearing tasks from a list.

Scratch is a block-based visual programming language that makes programming easy to understand. Using Scratch lists, variables, and event blocks, this project demonstrates how basic data management applications can be created in a simple and user-friendly way.





2. OBJECTIVE

The main objectives of this project are:

- To create a functional To-Do List using Scratch.
- To understand the use of Lists in Scratch.
- To implement keyboard controls for user interaction.
- To manage data dynamically (Add, Delete, Clear, View, Hide).
- To improve logical thinking and programming skills.

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3. SOFTWARE USED

- Scratch 3.0 (Online/Offline Editor)
- Platform: Web Browser / Scratch Desktop
- Operating System: Windows 11

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4. PROJECT DESCRIPTION

The To-Do List application allows users to:

- View the task list

- Hide the task list
- Add new tasks
- Delete specific tasks
- Clear the entire list

The program works using keyboard commands:

- Press **V** → View the list
- Press **H** → Hide the list
- Press **A** → Add items to the list
- Press **D** → Delete an item
- Press **C** → Clear the entire list

When the green flag is clicked, the program welcomes the user and displays instructions on how to use the application.

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5. WORKING PRINCIPLE

The application works using Scratch's **List data structure** named "To Do List".

5.1 When Green Flag Clicked

- The list is shown.
- A welcome message is displayed.
- Instructions are provided to the user.

5.2 Viewing the List (V Key)

- When the user presses "V", the list becomes visible.
- This allows the user to see all added tasks.

5.3 Hiding the List (H Key)

- When "H" is pressed, the list is hidden.
- This keeps the interface clean.

5.4 Adding Items (A Key)

- The program asks: "What would you like to add to your list?"
- The user types a task.
- The answer is added to the "To Do List".

- This process repeats until the user types “done”.
- A confirmation message is displayed.

5.5 Deleting an Item (D Key)

- The program asks for the item number to delete.
- It deletes the selected item from the list.
- A success message is shown.

5.6 Clearing the List (C Key)

- The program asks for confirmation.
- If the user enters “Y”, all items in the list are deleted.
- A confirmation message is displayed.

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6. ALGORITHM

1. Start the program.
2. When green flag is clicked:
    - Show list.
    - Display welcome message.
3. Wait for user key input.
4. If V is pressed → Show list.
5. If H is pressed → Hide list.
6. If A is pressed:
    - Ask user for task.
    - Add task to list.
    - Repeat until user enters “done”.
7. If D is pressed:
    - Ask for item number.
    - Delete that item.
8. If C is pressed:
    - Ask confirmation.
    - If answer = Y → Delete all items.
9. End.

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7. FEATURES

- Simple user interface
- Keyboard-based control
- Dynamic list management
- Confirmation before clearing
- User-friendly messages
- Real-time list updates

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8. ADVANTAGES

- Easy to use
- Beginner-friendly design
- Improves productivity
- Demonstrates practical use of Scratch lists
- Helps understand event-driven programming

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9. LIMITATIONS

- Data is not permanently stored (resets when project stops).
- No editing feature for tasks.
- No date or priority management.

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10. FUTURE IMPROVEMENTS

- Add task editing feature
- Add priority levels (High/Medium/Low)
- Add deadline feature
- Store tasks permanently using cloud variables
- Add graphical buttons instead of keyboard controls

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11. CONCLUSION

The To-Do List application developed using Scratch successfully demonstrates the use of lists, loops, conditions, and user interaction through keyboard inputs. The project enhances understanding of data handling and event-based programming concepts.

This project shows how simple programming logic can be used to build useful real-life applications. It is an effective beginner-level project that strengthens problem-solving and logical thinking skills.